Yao Ming (Brian) Chan

Personal Information

Email: brichan17@gmail.com
Date of birth: November 17th 1999
Place of birth: Ipoh, Malaysia
Nationality: Australian
Website: https://brichan17.github.io

Education

10/2023–Now **DPhil Mathematics**
University of Oxford
Expected graduation date: July 2027
Current thesis title: On the Cuntz semigroups of non-simple C*-algebras
Supervisor: Professor Stuart White

03/2021–11/2022 **MSc. Master of Science (Mathematics and Statistics)**
University of Melbourne
Major: Pure Mathematics
Thesis: Wedge product matrices and applications
Supervisor: Professor Arun Ram

03/2018–11/2020 **BSc. Bachelor of Science**
University of Melbourne
Major: Mathematics and Statistics
Specialisation: Pure Mathematics

Teaching

2025 **Class tutor for Functional Analysis I B4.1**

Organised and taught one set of classes (four classes total). Also played the role of the teaching assistant and marked the students' submitted work before each class.

2025 **Teaching assistant for Functional Analysis II B4.2**

Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class.

2023, 2024 **Teaching assistant for Functional Analysis I B4.1**

Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class.

Achievements

2021, 2022 **Mathematics and Statistics Masters Scholarship**

Three payments of $2000 corresponding to the first three semesters of the Master of Science degree in recognition of consistent academic achievement throughout the degree.

2018, 2019 **Dean's Honours List**

Awarded for an average mark which lies in the top 3% of students in the Bachelor of Science degree.

Preprints and publications

1. Y. Chan. *Wedge product matrices and orbits of principal congruence subgroups*, Lin Alg App. **696** (2024), 1–28.
2. Y. Chan. *The Cuntz semigroup of the graph C*-algebra of a finite graph*, preprint (arXiv: 2610.01915v2 [math.OA]), 2026.
3. Y. Chan, J. Curda, A. Shiner. *Classification of certain C*-algebras with a unique ideal*, currently in preparation.

Talks delivered

29/07/2026 **The Cuntz semigroup of a unital graph C*-algebra** (50 minute invited talk), UNSW, Sydney, Australia.

25/06/2026 **The Cuntz semigroup of a unital graph C*-algebra** (25 minute contributed talk), UK Operator Algebras Conference, Sabhal Mòr Ostaig, Isle

of Skye.

08/06/2026 **The Cuntz semigroup of a unital graph C*-algebra** (50 minute invited talk), Junior Functional Analysis Seminar, University of Oxford.

20/03/2026 **The Cuntz-Krieger relations in the context of the Cuntz semigroup** (15 minute contributed talk), Operator algebras in the south, Cardiff University.

11/09/2025 **Purely large extensions with purely infinite quotients** (20 minute contributed talk), Topological phenomena in the Cuntz semigroup, University of Oxford.

21/07/2025 **Cuntz semigroups and graph C*-algebras** (lightning talk), YMC*A, University of Southern Denmark, Odense.

04/07/2025 **Cuntz semigroups and graph C*-algebras** (lightning talk), UK Operator Algebras Conference, Queen's University Belfast.

03/04/2025 **Nuclear dimension, extensions and graph C*-algebras** (20 minute contributed talk), Young Functional Analysts' Workshop, University of Glasgow.

12/06/2024 **Nuclear dimension and graph C*-algebras** (lightning talk), UK Operator Algebras Conference, Newcastle University, Newcastle upon Tyne.

Research Visits and conferences

31/08/2026 - 04/09/2026 **Cuntz semigroups and dynamics** Chalmers University of Technology, Gothenburg, Sweden.

29/07/2026 - 31/07/2026 **Research visit** UNSW, Sydney, Australia.

21/06/2026 - 26/06/2026 **UK Operator Algebras Conference**, Sabhal Mòr Ostaig, Isle of Skye, Scotland.

19/03/2026 - 20/03/2026 **Operator algebras in the south**, Cardiff University, Cardiff, Wales.

10/09/2025 - 12/09/2025 **Topological phenomena in the Cuntz semigroup**, University of Oxford, Oxford, United Kingdom.

21/07/2025 - 25/07/2025 **Young Mathematicians in C*-algebras**, University of Southern Denmark, Odense, Denmark.

02/07/2025 - 04/07/2025 **UK Operator Algebras Conference**, Queen's University Belfast, Belfast, Northern Ireland.

02/04/2025 - 04/04/2025 **Young Functional Analysts' Workshop**, University of Glasgow, Glasgow, Scotland.

07/03/2025 - 07/03/2025 **Operator algebras in the south**, Cardiff University, Cardiff, Wales.

12/06/2024 - 14/06/2024 **UK Operator Algebras Conference**, Newcastle University, Newcastle upon Tyne, United Kingdom.

06/11/2023 - 10/11/2023 **Twinned Conference on C*-algebras and Tensor Categories**, The University of Edinburgh, Edinburgh, Scotland.

Skills

Coding experience with Python

2021 **Quantum chemistry project for the subject COMP90072**

Used NumPy to implement the Hückel method for modelling the energy levels, electron densities and bond orders of conjugated organic molecules. The Hartree-Fock method was also implemented to compute the orbital energies of small atoms with occupied 1s and 2s orbitals and diatomic molecules comprised of these atoms.

Languages

English (Native)

Professional Involvement

2025-2026 **Co-organiser** for the "Young Mathematicians in C*-algebras" conference, held at the University of Oxford in 2026.

Managed the email inbox for the conference, which had a total of 115 participants. Responded to various queries and requests from participants.

References

Professor Stuart White (DPhil supervisor)
Institute: University of Oxford
Email: stuart.white@maths.ox.ac.uk

Doctor Jennifer Pi
Institute: University of Oxford
Email: pi@maths.ox.ac.uk

Professor Aidan Sims
Institute: UNSW
Email: aidan.sims@unsw.edu.au